Computational Extension of OEIS Sequence A359813: Two New Terms at $n = 17$ and $n = 18$

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April 2026

Abstract

We extend OEIS sequence A359813, which counts primes less than 10^n having exactly one odd decimal digit, by computing two new terms: $a(17) = 40\,386\,401\,580$ and $a(18) = 190\,186\,145\,230$. The computation is based on a structural characterization of the candidate space (every prime in the sequence with at least two digits has its odd digit in the units position), a cascade of modular filters, and a deterministic Miller–Rabin test with twelve bases, all parallelised with OpenMP. The implementation is a single C17 source file (using the GCC/Clang `__uint128_t` extension for 128-bit arithmetic), no external dependencies beyond libc and OpenMP, and uses exclusively `uint64_t` and `__uint128_t` arithmetic. The search for $a(17)$ completes in approximately 2 h 20 min on a commodity workstation, with an independent re-computation using a different `mulmod` kernel producing an identical count at every individual prefix of the candidate space. The computation for $a(18)$ completes in approximately 12 h 49 min. Both results pass the Wilson heuristic sanity check within 1 %, and their ratio $a(18)/a(17)$ agrees with the Prime Number Theorem prediction to within 0.12 %.

2020 *Mathematics Subject Classification*: Primary 11A41; Secondary 11Y55, 11N05.
Key words and phrases: prime number, integer sequence, decimal digits, Miller–Rabin primality test, Montgomery multiplication, OpenMP, computational number theory.

1 Introduction

A prime p belongs to OEIS sequence A359813 [4] if $p < 10^n$ for some n and the decimal representation of p contains exactly one odd digit. The sequence $a(n)$ counts such primes as a function of n. For example, $a(1) = 3$ (the single-digit odd primes $3, 5, 7$; note that 2 has zero odd digits and is excluded), and $a(2) = 12$, as there are twelve primes less than 100 having a single odd digit:

$$3,\ 5,\ 7,\ 23,\ 29,\ 41,\ 43,\ 47,\ 61,\ 67,\ 83,\ 89.$$

Yang [9] introduced the sequence in January 2023 and computed its initial terms. Robert G. Wilson v extended the sequence to $a(16) = 8\,600\,149\,593$ (February 2023). The sequence carries the keyword `more`, signalling interest in further computational extension.

The broader question of the existence and distribution of primes with prescribed digit constraints has received substantial theoretical attention, most notably Maynard's proof [1] that there are infinitely many primes avoiding any fixed digit. The present sequence imposes a complementary constraint (exactly one odd digit), which is more restrictive and is studied here from a purely computational perspective.

The computational difficulty has two components. First, the number of candidates grows as $16 \cdot 5^{n-2}$ by the structural Lemma 1 below, which for $n = 17$ is approximately 4.88×10^{11} and for $n = 18$ is approximately 2.44×10^{12}. Second, each surviving candidate requires a primality

test on an integer of up to 18 decimal digits, comfortably within a 64-bit unsigned integer but sufficient to require care in the choice of witnesses for a deterministic Miller–Rabin test.

The present work addresses both components through a structural reduction of the enumeration space (Section 2), a cascade of modular filters followed by deterministic Miller–Rabin with twelve bases (Section 3), and an OpenMP implementation (Section 3) running on a commodity 16-thread workstation. The main results are as follows.

1. $a(17) = 40\,386\,401\,580$ (Theorem 7).
2. $a(18) = 190\,186\,145\,230$ (Theorem 8).
3. A structural lemma forcing the odd digit of every prime with at least two digits in the sequence to coincide with the units digit (Lemma 1).

Both new terms pass the Wilson heuristic sanity check within 1 %, and are mutually consistent through the Prime Number Theorem ratio test (Section 5).

1.1 Notation

Throughout, p denotes a prime, n a positive integer, and $\pi(x)$ the prime-counting function. We write $[L, R)$ for half-open integer intervals. For any positive integer m, we denote by $\mathrm{digs}(m)$ the decimal digits of m without leading zeros, and by $|m|_{10}$ the number of such digits. The candidate space for $a(n)$ is denoted $\mathcal{C}_n$ (Definition 4).

2 Mathematical framework

2.1 Structural characterization of candidates

We begin with a simple observation that dramatically reduces the enumeration space.

Lemma 1. *Let $p > 2$ be a prime whose decimal representation contains exactly one odd digit. Then that odd digit must be the units digit of p.*

Proof. In base 10, an integer m is odd if and only if its units digit is odd, since $m \equiv d_0 \pmod 2$ where d_0 is the units digit. Since $p > 2$ is prime, p is odd, hence its units digit d_0 is odd. By hypothesis p contains exactly one odd digit, so the single odd digit must be d_0, i.e. the units digit. □

Remark 2 (Edge case $p = 2$)**.** The prime $p = 2$ has the decimal representation "2", which contains *zero* odd digits. Hence $p = 2$ never belongs to the sequence, and Lemma 1 applies to every prime in the sequence.

Remark 3. Lemma 1 is an identity, not a heuristic. It holds for every prime $p > 2$ whose decimal representation contains exactly one odd digit, and it is this exact structural fact that drives the enumeration.

Definition 4 (Candidate space)**.** For $n \geq 2$, define

$$\mathcal{C}_n := \{2, 4, 6, 8\} \times \{0, 2, 4, 6, 8\}^{n-2} \times \{1, 3, 7, 9\},$$

interpreted as the set of n-digit integers of the form $d_{n-1} \cdot 10^{n-1} + d_{n-2} \cdot 10^{n-2} + \cdots + d_0$, where the leading digit $d_{n-1} \in \{2, 4, 6, 8\}$, the interior digits $d_i \in \{0, 2, 4, 6, 8\}$ for $1 \leq i \leq n-2$, and the units digit $d_0 \in \{1, 3, 7, 9\}$. For $n = 1$, we set $\mathcal{C}_1 := \{1, 3, 5, 7, 9\}$.

The leading digit is restricted to $\{2,4,6,8\}$ because a prime of $n \geq 2$ digits cannot have leading digit 0, and because the only primes with leading digit $1,3,5,7,9$ and exactly one odd digit are the single-digit primes; these are handled separately for $n = 1$. It follows immediately from Lemma 1 that every prime counted by $a(n)$ with at least two digits lies in $\mathcal{C}_n$.

Proposition 5. *For $n \geq 2$, we have $|\mathcal{C}_n| = 16 \cdot 5^{n-2}$.*

Proof. Immediate from Definition 4: $|\{2,4,6,8\}| = 4$, $|\{0,2,4,6,8\}^{n-2}| = 5^{n-2}$, $|\{1,3,7,9\}| = 4$, and $4 \cdot 5^{n-2} \cdot 4 = 16 \cdot 5^{n-2}$. □

For $n = 17$, Proposition 5 gives $|\mathcal{C}_{17}| = 488\,281\,250\,000$; for $n = 18$, $|\mathcal{C}_{18}| = 2\,441\,406\,250\,000$. Compared with the naive space $[0, 10^n)$, the structural reduction is a factor of exactly

$$\frac{10^n}{16 \cdot 5^{n-2}} = \frac{2^n \cdot 5^n}{2^4 \cdot 5^{n-2}} = 25 \cdot 2^{n-4},$$

i.e. a factor of $25 \cdot 2^{13} = 204\,800$ at $n = 17$.

2.2 Modular filter

A candidate $c \in \mathcal{C}_n$ is not necessarily prime. The vast majority of candidates are composite and can be eliminated by a cascade of small-prime divisibility tests before the costly Miller–Rabin stage. We apply two levels of modular filter.

Filter by 3. The value of $c \bmod 3$ depends only on the sum of its digits modulo 3. For a fixed choice of the $n-1$ leading digits, the units digit $u \in \{1,3,7,9\}$ is constrained by the requirement that $c \not\equiv 0 \pmod 3$ (except for the trivial case $c = 3$). A direct computation gives, writing s for the sum of the leading digits:

$$\begin{aligned} s \equiv 0 \pmod 3 &\implies u \in \{1,7\}, \\ s \equiv 1 \pmod 3 &\implies u \in \{1,3,7,9\}, \\ s \equiv 2 \pmod 3 &\implies u \in \{3,9\}. \end{aligned}$$

This filter eliminates, to excellent approximation, 1/3 of the candidates. To see why: the units digits $\{1,3,7,9\}$ satisfy $1 \equiv 7 \equiv 1 \pmod 3$ and $3 \equiv 9 \equiv 0 \pmod 3$, so exactly two of the four units-digit choices are $\equiv 0$ and two are $\equiv 1 \pmod 3$. For a fixed prefix sum $s \equiv 0 \pmod 3$ exactly 2 of 4 units digits survive; for $s \equiv 1 \pmod 3$ all 4 survive (since none are $\equiv 2$); and for $s \equiv 2 \pmod 3$ exactly 2 of 4 survive. If prefix digit sums were uniformly distributed modulo 3 over $\mathcal{C}_n$, the average survival rate would be $(2/4 + 4/4 + 2/4)/3 = 2/3$, i.e. the filter would eliminate exactly 1/3.

The distribution of prefix sums modulo 3 is, however, only asymptotically uniform, not exactly so for finite n. Each interior digit from $\{0,2,4,6,8\}$ contributes a residue drawn from the distribution $(0,1,2) \mapsto (2/5, 1/5, 2/5)$, whose associated circulant transition matrix has second eigenvalue of modulus 1/5; after the $n-2$ interior digits the deviation from uniform is bounded by $(1/5)^{n-2}$. For the leading digit from $\{2,4,6,8\}$ the residues $(0,1,2)$ occur with frequencies $(1/4, 1/4, 2/4)$, introducing an initial non-uniformity that is then suppressed exponentially by the interior digits. For $n = 2$ (where there are no interior digits) the elimination rate is $3/8 = 37.5\,\%$, not 1/3; for $n \geq 4$ the deviation from 1/3 is smaller than $(1/5)^{n-2}/3$, and for $n = 17$ it is below 10^{-10}, negligible for all computational purposes. In the implementation it is encoded as a three-entry bitmask on the residue class of the leading-digit sum, applied inside the enumeration loop.

Filter by small primes. Let

$$\mathcal{P}_{\text{small}} := \{7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47\}.$$

For each prime $p \in \mathcal{P}_{\text{small}}$ we precompute a table contrib[p][pos][digit] giving the contribution digit $\cdot\, 10^{\text{pos}} \bmod p$ for every possible digit and position. The residue of the candidate modulo each small prime is maintained incrementally as the enumeration advances, and the candidate is discarded as soon as one of these residues vanishes. A minor implementation detail requires the guard $c > p$, to avoid discarding the prime p itself when it coincides with a candidate (relevant only for $n = 2$).

Empirically, this filter eliminates approximately 48 % of the candidates that survive the filter by 3, leaving roughly 35 % of the original space to be processed by the Miller–Rabin test.

2.3 Primality test

After filtering, each surviving candidate $c < 10^{18}$ is tested by deterministic Miller–Rabin with the twelve bases

$$\mathcal{B}_{12} := \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37\},$$

which constitute a deterministic witness set for every odd composite n below

$$3\,317\,044\,064\,679\,887\,385\,961\,981 \;\approx\; 3.3 \times 10^{23},$$

as established by Sorenson and Webster [7] and recorded as OEIS sequence A014233 [3].

Remark 6 (Why twelve bases and not seven)**.** A common choice in the literature is the set of seven bases $\{2, 3, 5, 7, 11, 13, 17\}$, following Jaeschke [2]. This set is deterministic only for $n < 341\,550\,071\,728\,321 \approx 3.4 \times 10^{14}$; for the targets $a(17)$ and $a(18)$, which require primality certification of candidates up to 10^{18}, this set is *not* deterministic. We use the twelve-base set throughout.

The Miller–Rabin kernel uses Montgomery form: for each candidate c, the Montgomery parameters $c^{-1} \bmod 2^{64}$ and $2^{128} \bmod c$ are computed once, and every modular multiplication is then performed as a Montgomery reduction. A fallback kernel using native `__uint128_t` multiplication is also implemented; this is used only for the cross-check described in Section 5.

3 Implementation

3.1 Enumeration

For $n \geq 4$, the candidate space $\mathcal{C}_n$ is partitioned into 100 subspaces indexed by the first three digits of the candidate. The first three digits lie in $\{2, 4, 6, 8\} \times \{0, 2, 4, 6, 8\}^2$, yielding $4 \times 5 \times 5 = 100$ combinations. Each subspace contains $4 \cdot 5^{n-4}$ raw candidates; after the mod-3 filter (which eliminates approximately 1/3 of candidates in aggregate) each subspace contributes on average $(8/3) \cdot 5^{n-4} \approx 2.67 \cdot 5^{n-4}$ survivors.

The enumeration within a subspace is a pure digit-tree walk: the middle digits are iterated through all 5^{n-4} combinations in a Gray-code-like order that maintains, incrementally, the partial decimal value of the candidate, the sum of digits modulo 3, and the residues modulo each prime in $\mathcal{P}_{\text{small}}$. No intermediate candidate is ever materialized in memory beyond a constant-size working set.

For $n \in \{1, 2, 3\}$, the enumeration is handled by direct cases with fewer than 100 candidates, without the prefix partitioning.

3.2 Parallelisation

The 100 prefix-level subspaces form the natural unit of parallelism: they are mutually disjoint, roughly balanced in workload, and require no shared state during execution. The OpenMP directive

```
#pragma omp parallel for schedule(dynamic,1) reduction(+:total)
```

assigns subspaces to threads dynamically. On the target hardware (AMD Ryzen 9 7940HS, 8 cores, 16 threads via SMT2), a scaling experiment on $a(12)$ yielded the speedup shown in Table 1.

Threads	Time (s)	Speedup	Efficiency
1	34.521	1.00×	100.0 %
2	17.296	2.00×	99.8 %
4	8.759	3.94×	98.5 %
8	4.478	7.71×	96.4 %
16	2.768	12.47×	77.9 %

Table 1: OpenMP scaling on $a(12)$. Efficiency drops at 16 threads because the Ryzen 9 7940HS has only 8 physical cores; the last 8 threads share execution resources via SMT2.

3.3 Checkpointing

For long-running campaigns such as $a(18)$, the implementation writes a binary checkpoint after each of the 100 subspaces completes. The checkpoint records the completed prefix bitmap, the per-prefix partial count, and a CRC-32 integrity tag, and is written atomically via `rename()` on a temporary file. A run invoked with `--resume` loads the checkpoint and skips the completed prefixes. The checkpoint granularity of approximately 1 % of the total workload ensures that no more than a few minutes of computation is ever at risk of loss in case of interruption.

3.4 Hardware and software environment

All runs reported in this paper were executed on the same workstation:

- CPU: AMD Ryzen 9 7940HS (8 cores, 16 threads, max boost 5.26 GHz);
- RAM: 64 GB DDR5-5600;
- OS: Linux Mint 22.3 (kernel 6.14);
- Compiler: GCC 13.3.0;
- Compile flags: `-std=c17 -fopenmp -O3 -march=native -DNDEBUG`;
- Note: the implementation uses `__uint128_t`, a GCC/Clang compiler extension not part of ISO C17; the source is therefore correctly described as GNU C17 rather than strictly ISO C17;
- Peripheral tools: Python 3.12, SymPy 1.14 (for cross-checks).

4 Computational results

4.1 Oracle verification

Before running the campaigns for $a(17)$ and $a(18)$, the entire implementation was validated against the OEIS b-file for $a(1), \ldots, a(16)$, using the same code and the same twelve-base Miller–Rabin kernel. At 16 OpenMP threads, the full oracle suite completes in approximately 43 minutes and matches the b-file on every term with an exact count.

Crucially, the oracle test was also run at 1 thread and yielded the identical counts at every layer, providing strong empirical evidence against harmful race conditions in the parallel enumeration (see [6] for the representative validation logs).

4.2 First new term

Theorem 7. $a(17) = 40\,386\,401\,580$.

Proof sketch. The result follows from the exhaustive enumeration of $\mathcal{C}_{17}$ (Definition 4, cardinality 488 281 250 000), with primality determined by deterministic twelve-base Miller–Rabin (Section 2.3). The enumeration is complete in the sense that every integer in $[10^{16}, 10^{17})$ whose decimal representation has exactly one odd digit is visited, by Lemma 1. The result is the cumulative count

$$a(17) \;=\; a(16) \;+\; \#\{\, p \in \mathcal{C}_{17} : p \text{ prime} \,\},$$

with $a(16) = 8\,600\,149\,593$ taken from the OEIS b-file and the layer count computed from the present campaign. □

This computation was repeated with a second, arithmetically independent implementation of modular multiplication: native `__uint128_t` reduction instead of Montgomery form. Both runs produced the identical result; see Section 5.

4.3 Second new term

Theorem 8. $a(18) = 190\,186\,145\,230$.

Proof sketch. Analogous to Theorem 7, with the cumulative relation

$$a(18) \;=\; a(17) \;+\; \#\{\, p \in \mathcal{C}_{18} : p \text{ prime} \,\},$$

where $a(17)$ is taken from Theorem 7. □

4.4 Wilson heuristic sanity check

Wilson [8] observed the empirical relation

$$a(n) \approx \frac{\pi(10^n)}{2^{n-1}}. \tag{1}$$

Using the known values of $\pi(10^{17})$ and $\pi(10^{18})$ [5], equation (1) yields

$$\frac{\pi(10^{17})}{2^{16}} = 40\,032\,305\,262, \qquad \frac{\pi(10^{18})}{2^{17}} = 188\,750\,871\,946.$$

n	$a(n)$	Wilson estimate	Relative scatter
17	40 386 401 580	40 032 305 262	0.8768 %
18	190 186 145 230	188 750 871 946	0.7547 %

Table 2: Wilson sanity check for the two new terms. Both values pass the hard-coded 2 % threshold embedded in the implementation, which aborts the run with a fatal error when exceeded. Relative scatter is defined as $|a(n) - \text{estimate}|/a(n)$ (denominator is the observed value; using the estimate as denominator gives values differing by less than 0.01 %).

Run	Backend	Threads	Wall-time	Result
Oracle $a(1)-a(16)$	Montgomery, 12 bases	16	42 min 54 s	b-file match
$a(17)$, primary	Montgomery, 12 bases	16	2 h 20 min	40 386 401 580
$a(17)$, cross-check	`u128`, 12 bases	16	4 h 12 min	40 386 401 580
$a(18)$, primary	Montgomery, 12 bases	14	12 h 49 min	190 186 145 230

Table 3: Summary of the four principal runs. The $a(18)$ run was performed at 14 threads rather than 16 to reduce sustained CPU temperature from 92 °C (observed on the $a(17)$ run at 16 threads) to 80 °C, at a performance cost of approximately 8 %.

4.5 Campaign summary

Table 3 summarizes the four principal runs performed for this work.

The total campaign time for the two new terms is approximately 20 hours of wall-clock time, spread over three days, including the kernel cross-check.

5 Independent validation

The published values of $a(17)$ and $a(18)$ are supported by multiple validation layers; their scope differs between the two new terms.

For $a(17)$: an arithmetically independent kernel cross-check (Section 5.1), a partial enumerator cross-check against SymPy on prefixes of $a(14)$ (Section 5.2), and internal consistency with PNT and Wilson heuristic. For $a(18)$: internal consistency with $a(17)$ via the PNT ratio test and with the Wilson heuristic. The algorithm's correctness, demonstrated for $a(17)$ and by the oracle match on $a(1)$–$a(16)$, provides the foundational guarantee that the same code path applied to $a(18)$ is free of systematic error.

5.1 Kernel cross-check on $a(17)$

The primary campaign for $a(17)$ used Montgomery-form modular multiplication. The cross-check campaign used `__uint128_t` native multiplication, which is an arithmetically distinct code path with different carry-propagation paths, different register usage, and different compiler-generated assembly. Both runs produced $a(17) = 40\,386\,401\,580$, with identical counts at every individual prefix. The per-prefix partial counts (all 100 prefixes) also matched, providing stronger evidence than a single final-total comparison.

The observed wall-time ratio between the two kernels (4.20 h vs. 2.34 h) is 1.796. Combined with the isolated kernel ratio of 1.90 measured in a separate benchmark, this implies that the Miller–Rabin kernel accounts for approximately 88 % of the total pipeline time on the Ryzen 9 7940HS.

5.2 Enumerator cross-check against SymPy

A third, independent implementation was written in Python using `sympy.isprime` as the primality oracle. This implementation replicates the enumeration of $\mathcal{C}_n$ exactly (same prefix decoding, same modular filters, same mask by 3) but differs in every other respect: interpreted language, different library, and a BPSW-based primality test instead of Miller–Rabin. For $n < 2^{64}$, the `sympy.isprime` function is conjecturally deterministic (its correctness rests on the Baillie–PSW conjecture, which remains unproven but has no known counterexample below 2^{64}).

For computational feasibility, the cross-check was performed on $a(14)$ rather than $a(17)$, and on individual prefixes rather than the whole sequence. Table 4 summarises the three prefixes selected, chosen to span all four possible values of the leading digit and the two extremes of the middle-digit distribution.

pid	digits	C (MR-12)	Python (BPSW)	match
0	$(2,0,0)$	3 193 483	3 193 483	✓
75	$(8,0,0)$	3 049 153	3 049 153	✓
99	$(8,8,8)$	3 059 841	3 059 841	✓

Table 4: Cross-check of the enumerator on three prefixes of $a(14)$. The C implementation uses deterministic twelve-base Miller–Rabin in Montgomery form; the Python implementation uses `sympy.isprime`, a BPSW-based primality test conjecturally deterministic for $n < 2^{64}$ (Baillie–PSW conjecture; no known counterexample). Counts are exact and identical on all three prefixes.

As a secondary sanity check, the ratio of densities predicted by the Prime Number Theorem between the smallest-magnitude prefix ($2 \cdot 10^{13}$) and the largest-magnitude prefix ($8.88 \cdot 10^{13}$) is

$$\frac{1/\ln(2 \cdot 10^{13})}{1/\ln(8.88 \cdot 10^{13})} = \frac{\ln(8.88 \cdot 10^{13})}{\ln(2 \cdot 10^{13})} = \frac{32.117}{30.627} \approx 1.0487,$$

while the observed ratio is $3\,193\,483/3\,059\,841 = 1.0437$, a deviation of approximately 0.48 %.

5.3 Internal consistency of $a(17)$ and $a(18)$

The Prime Number Theorem predicts

$$\frac{a(n+1)}{a(n)} \approx \frac{\pi(10^{n+1})/2^n}{\pi(10^n)/2^{n-1}} = \frac{1}{2} \cdot \frac{\pi(10^{n+1})}{\pi(10^n)}.$$

For $n = 17$, this yields $\frac{1}{2} \cdot \pi(10^{18})/\pi(10^{17}) = 4.7150$, while the observed value is $a(18)/a(17) = 4.7092$, a deviation of 0.12 %. This is an independent check that both new terms are internally consistent with each other, not only individually consistent with Wilson's heuristic.

6 Heuristic discussion

Wilson's relation (1) can be motivated by a density argument, though a rigorous derivation remains open (Open Problem 3).

A naive starting point is to treat the $|\mathcal{C}_n| = 16 \cdot 5^{n-2}$ candidates as uniformly sampled from $[1, 10^n)$. Applying the global prime density $\pi(10^n)/10^n$ gives an expected count of

$$16 \cdot 5^{n-2} \cdot \frac{\pi(10^n)}{10^n} = \frac{16}{25} \cdot \frac{\pi(10^n)}{2^n} \approx 0.64 \cdot \frac{\pi(10^n)}{2^n}.$$

This underestimates the observed values by a factor of roughly 3. The discrepancy is expected: the global density treats every integer as equally likely to be prime, ignoring that every element of $\mathcal{C}_n$ ends in a digit from $\{1,3,7,9\}$, and is therefore coprime to both 2 and 5. Since every prime $p > 5$ ends in one of these four digits, which account for 4/10 of all integers, the prime density *among* $\{1,3,7,9\}$-ending integers is approximately

$$\frac{\pi(10^n)}{(4/10) \cdot 10^n} = \frac{5}{2} \cdot \frac{\pi(10^n)}{10^n}.$$

Substituting this refined density yields

$$16 \cdot 5^{n-2} \cdot \frac{5}{2} \cdot \frac{\pi(10^n)}{10^n} = \frac{8}{5} \cdot \frac{\pi(10^n)}{2^n} = 1.6 \cdot \frac{\pi(10^n)}{2^n},$$

which is much closer to, but still about 20 % below, Wilson's empirical formula $2 \cdot \pi(10^n)/2^n = \pi(10^n)/2^{n-1}$. The remaining discrepancy is left as an open question (see Open Problem 3); its

sign and magnitude suggest a further positive arithmetic bias in the density of primes within $\mathcal{C}_n$ relative to the global density among all $\{1, 3, 7, 9\}$-ending integers, but quantifying this rigorously is non-trivial.

In summary, the density argument provides qualitative support for a formula of the form $c \cdot \pi(10^n)/2^n$ with $c \sim 1$–2, and Wilson's empirical choice $c = 2$ (i.e. $a(n) \approx \pi(10^n)/2^{n-1}$) is best regarded as an observed regularity rather than a derived result.

Conjecture 9. *The sequence $\{a(n)\}_{n\geq 1}$ is strictly increasing and tends to infinity.*

Conjecture 9 is strongly suggested by Wilson's heuristic, which predicts $a(n) \sim 5^n/(n \cdot \text{const})$, a quantity tending to infinity. However, neither the strict monotonicity nor the divergence follows rigorously from the heuristic, which is itself unproved. The non-strict monotonicity $a(n+1) \geq a(n)$ is trivially true, since every prime counted by $a(n)$ is also counted by $a(n+1)$; the conjecture asserts strict inequality for all n, which requires the existence of at least one prime with exactly one odd digit and exactly $n+1$ decimal digits for every $n \geq 1$.

7 Open problems

Open Problem 1. Compute $a(19)$. The candidate space has cardinality $16 \cdot 5^{17} \approx 1.22 \times 10^{13}$, a factor of 5 larger than that of $a(18)$; the expected wall-time at 14 threads on the present hardware is approximately 68 hours. The ratio $a(19)/a(18)$ is predicted to be approximately $\frac{1}{2} \cdot \pi(10^{19})/\pi(10^{18}) \approx 4.73$.

Open Problem 2. Prove Conjecture 9.

Open Problem 3. Derive a rigorous error bound for Wilson's heuristic (1). The empirical evidence at $n \leq 18$ suggests an error of order $O(a(n)/\ln(10^n))$, but a formal statement appears non-trivial.

Open Problem 4. By Lemma 1, every prime in the sequence has its sole odd digit in the units position, so no genuinely distinct prime sequence arises by allowing the odd digit at other positions. A natural extension is to study the sequence $b(n) := \#\{m \in \mathbb{Z} : 1 \leq m < 10^n,\ \omega_{\text{odd}}(m) = 1\}$, where $\omega_{\text{odd}}(m)$ denotes the number of odd decimal digits of m, counting without regard to position. Unlike $a(n)$, the sequence $b(n)$ counts all positive integers below 10^n with exactly one odd digit, including composites; here the odd digit may occupy any position. Determine the growth rate of $b(n)$ and compare it with $a(n)$.

8 Data and code availability

The complete source code, the extended b-file for $a(1), \ldots, a(18)$, the main campaign logs, and the full methodological documentation are available at the GitHub repository [6].

The single-file C17 implementation (`A359813.c`), together with the shell scripts for reproducing the oracle test, the $a(17)$ campaign, and the $a(18)$ campaign, is sufficient to reproduce every numerical result in this paper on any 64-bit Linux machine with GCC 11 or newer.

Acknowledgements

The author thanks Zhining Yang for creating OEIS sequence A359813, and Robert G. Wilson v for extending it to $a(16)$ and for the heuristic relation that provided the sanity check used throughout this work.

AI use disclosure

This work was carried out with substantial support from large language models (Claude and ChatGPT), used as a research assistant throughout the full workflow: mathematical analysis of the candidate space, design of the computational strategy, code generation in C17, independent peer review of the generated code, and composition of this manuscript. Every quantitative claim in this paper was verified computationally; all mathematical statements were checked by the author before inclusion. The author takes full scientific responsibility for the content. The exact model versions and API identifiers used will be recorded in the repository at [6].

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